

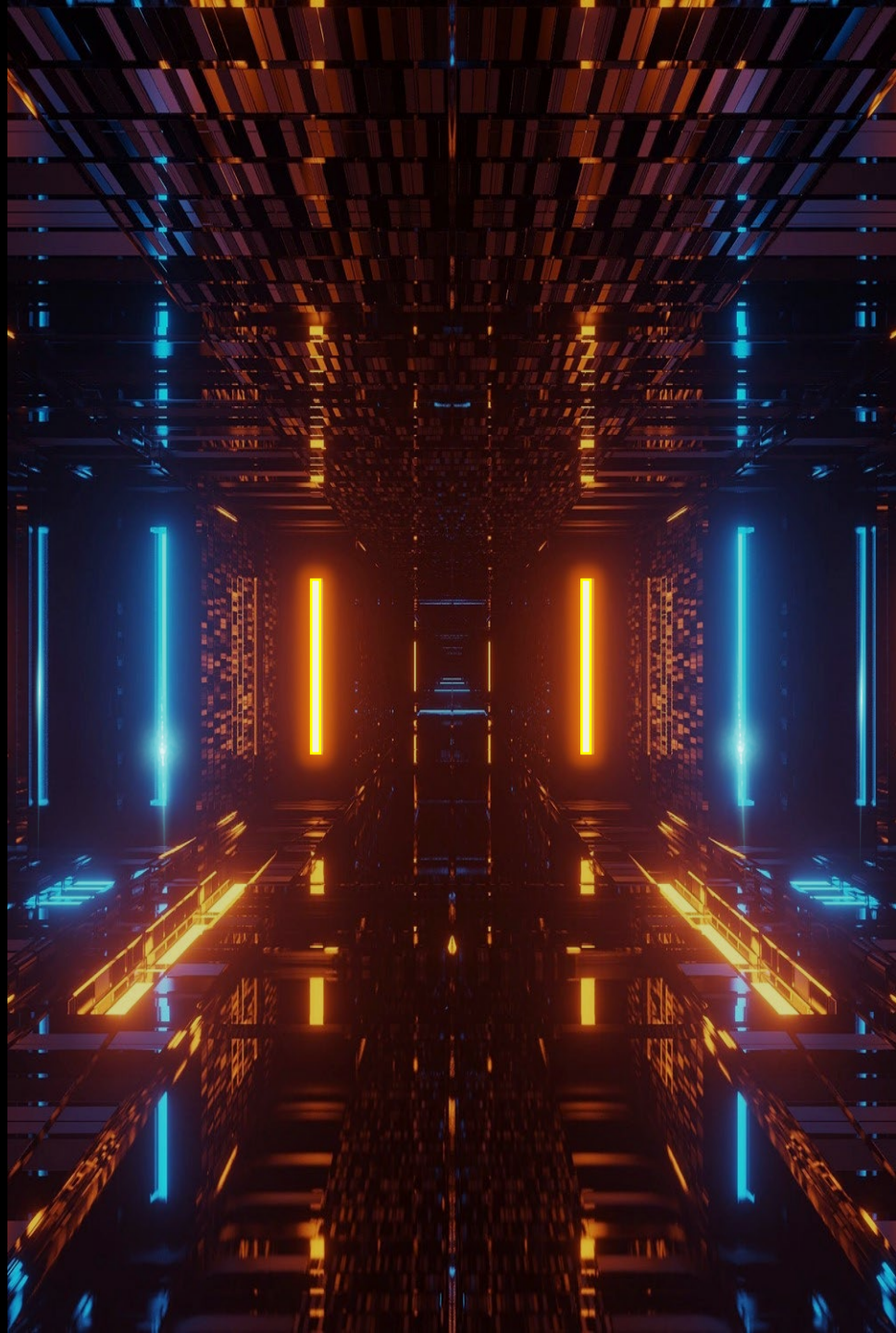
2026 04.28 —————> 06.19

数据科学与大数据 技术的数学基础

Mathematical Foundations of
Data Science

罗霄

317834051@qq.com



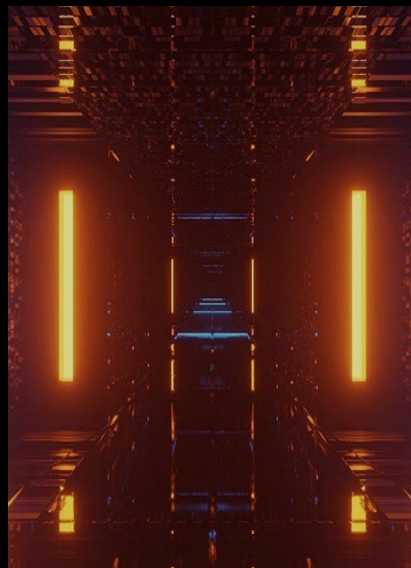
数据科学与大数据技术的数学基础

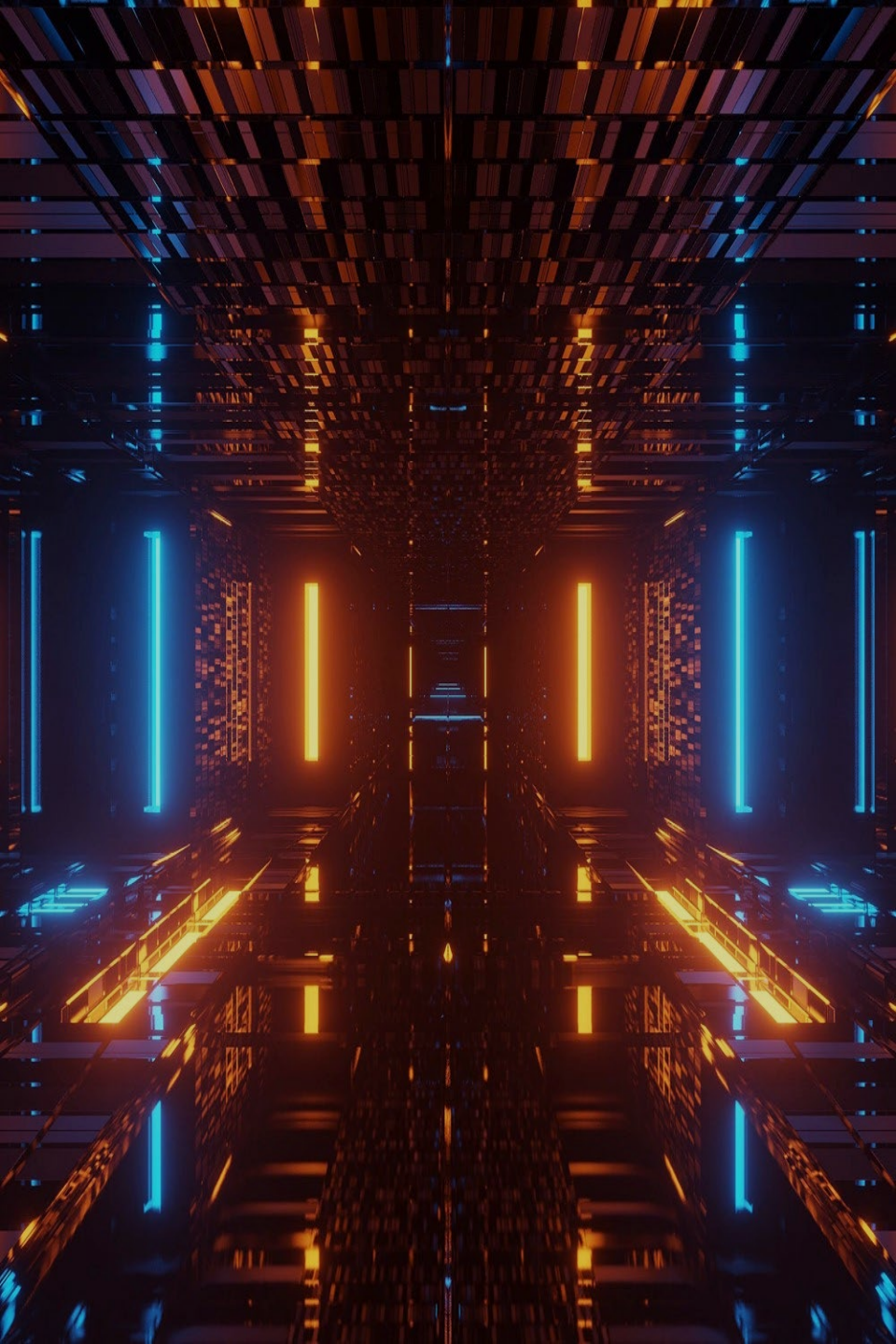
Mathematical Foundations of Data Science

/04

主成分分析

Principal Component
Analysis (PCA)





Contents

1. Data visualization in low-dimensional space
2. Principal Component Analysis (PCA)
3. Solving steps
4. Why
5. Advantages and disadvantages of PCA

4.1 Data visualization in low-dimensional space

If you are a junior high school class teacher and need to analyze your students' scores in various subjects

学生编号	语文	数学	物理	化学
1	90	140	99	100
2	90	97	88	92
3	90	110	79	83
.....				

principal components

math as the **first principal component**, because math scores pull the widest

First, assume that these subject scores are **unrelated**, the scores in one subject are not related to the other subjects.

There are too many data and couldn't find the principal components of this data set directly

- Then for higher dimensional data, does the **distribution** can be imagined?
- Even if you can describe the distribution, **how can you find the axes** of these principal components precisely?
- How can you measure **how much information** of the whole data is **accounted for by the principal components** you extracted?

So, we have to use the **principal component analysis**.

4.1 Data visualization in low-dimensional space



In many areas of research and applications, it is often necessary to collect large amounts of data for analysis and modeling purposes.

The characteristics reflecting the characteristics of the sample may be **high-dimensional vectors** (e.g., we analyze the scores of multiple courses), and **large samples with multiple variables** will undoubtedly provide rich information for research and applications.

But it also increases the workload of data collection, and more importantly, in most cases, **many variables may be correlated with each other**, thus increasing the complexity of the problem analysis and at the same time, the analysis inconvenience.

If each dimension is analyzed separately, the **analysis** (the conclusions obtained) is often **isolated** rather than integrated. **Blindly reducing the characteristic dimensions** can **lose a lot of information** and tend to produce **wrong conclusions**.

Our goal is to **reduce the dimensionality without excessive loss of information** in the original sample, **which features can be ignored?** Is there **some composite feature** or **composite indicator** that can represent several feature variables?



While **reducing the number of features**, **the loss of information** contained in the original indicators **was reduced** as much as possible in order to achieve a comprehensive analysis of the collected data.

Since there are certain correlations among the variables, it is possible to conflate the various types of information present in each variable separately **using fewer composite indicators**.

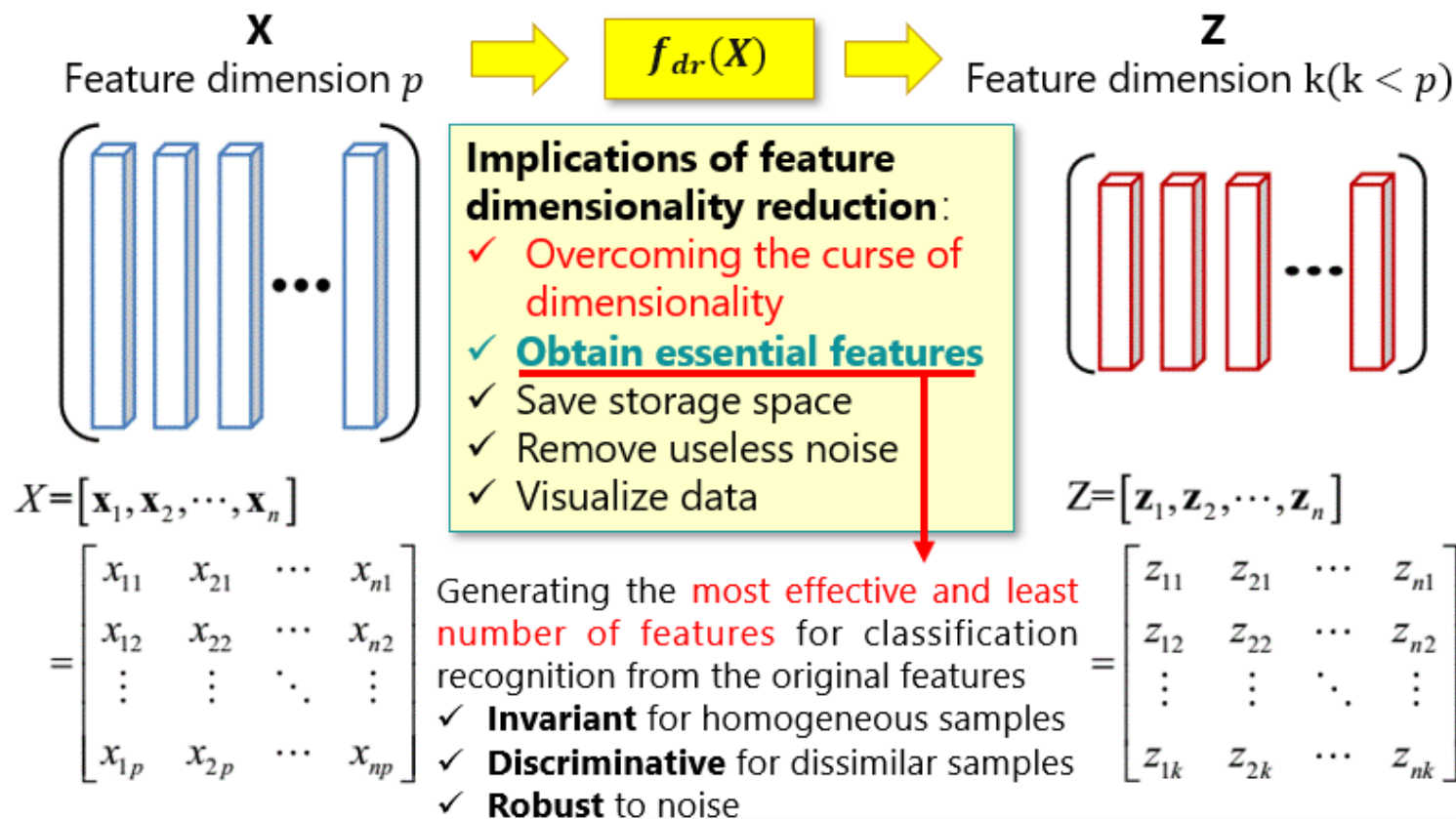
Principal component analysis and factor analysis are this methods of dimensionality reduction.

PS Dimensionality Reduction

2.5 Curse of Dimensionality: dimensionality reduction

The Curse of Dimensionality

Dimensionality Reduction

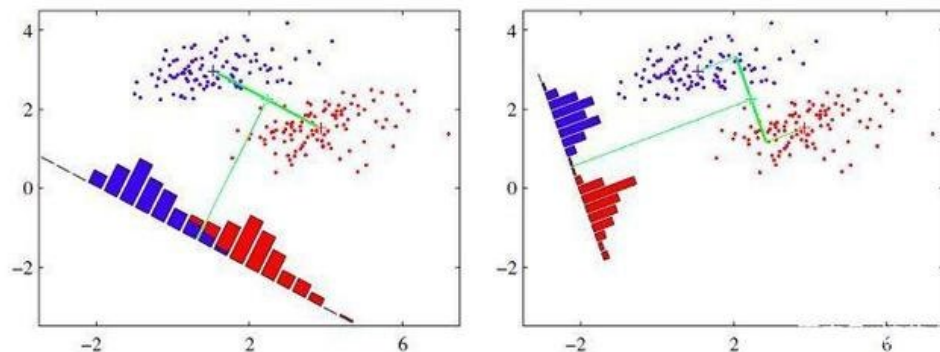


Common methods for dimensionality reduction.

- Feature selection
- Feature transformation

PS Dimensionality Reduction

Linear dimensionality reduction: the low-dimensional data can **maintain the linear relationship** between the high-dimensional data points, mainly including PCA, LDA, LPP (traditional linear techniques mainly want to keep dissimilar points separated in the low-dimensional representation)



Choose the right projection angle, the intraclass variation is small and the interclass variation is large

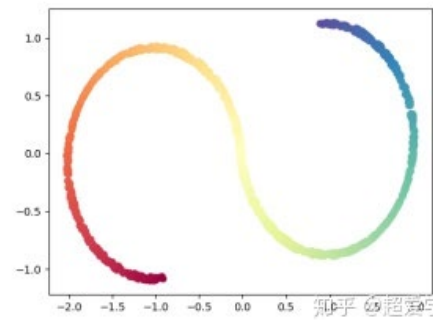
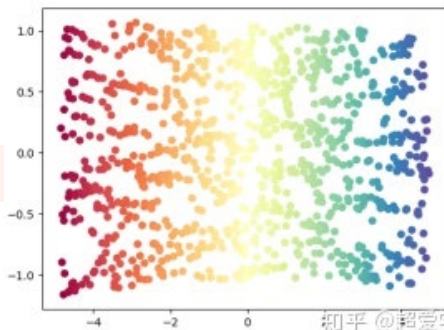
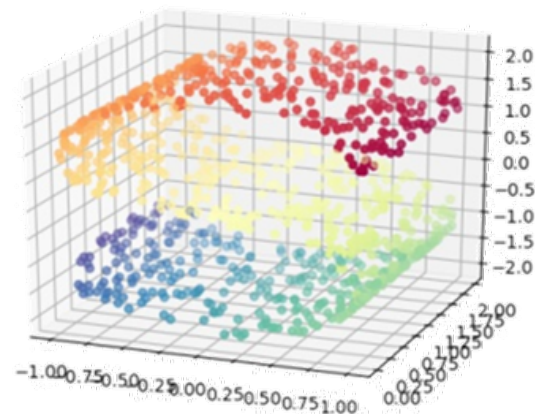
Linear methods are fast and low in complexity, but poor in dimensionality reduction for complex data

Nonlinear dimensionality reduction:

① Kernel function-based approach, such as KPCA

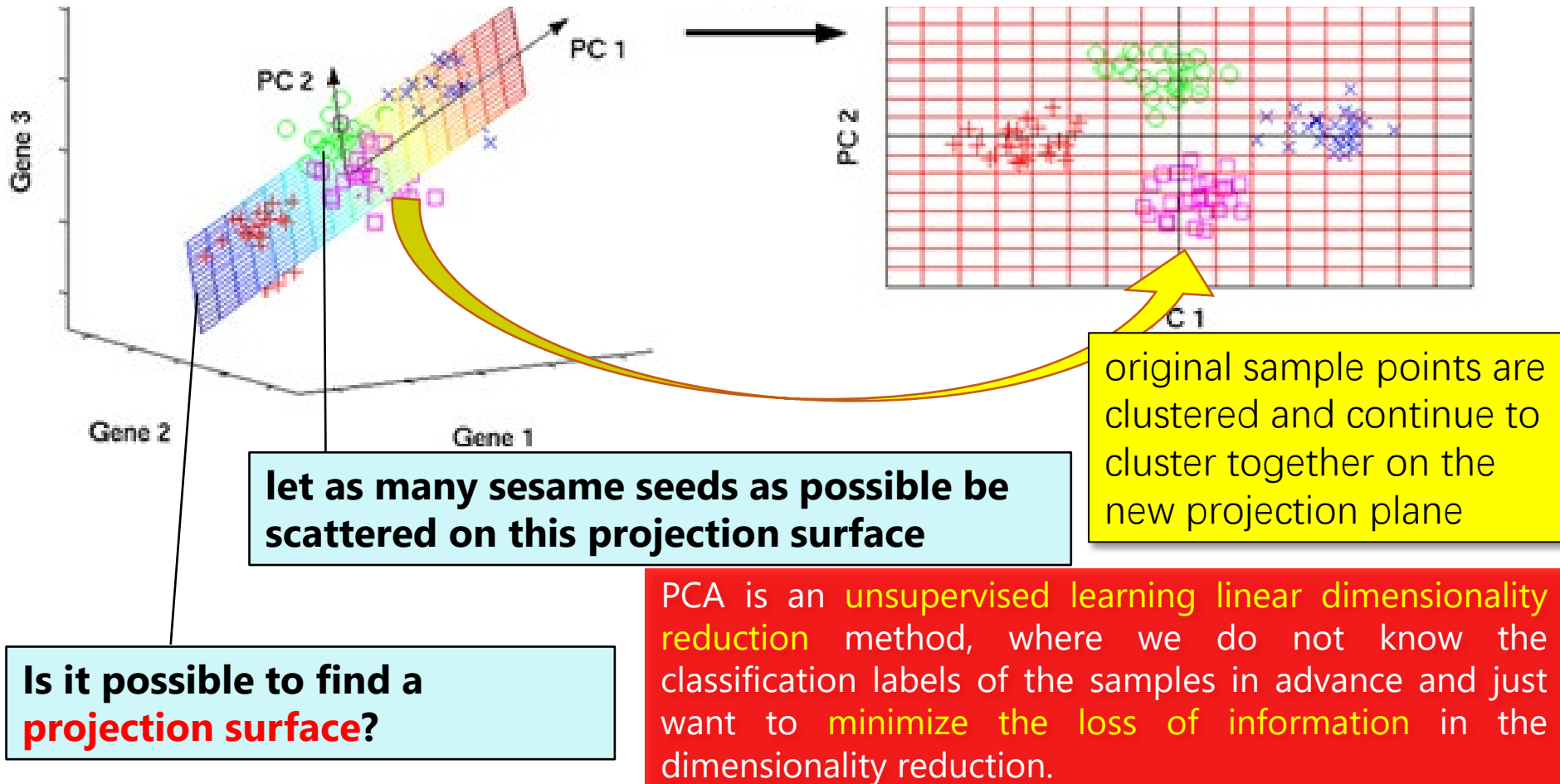
② manifold learning: recovering the low-dimensional manifold structure from high-dimensional data and deriving the corresponding embedding mapping, with Isomap, LLE, Laplacian Eigenmaps, LTSA, MVU

(nonlinear techniques mainly **maintain the local structure of the data**)



4.2 Principal Component Analysis (PCA)

- ✓ The previous page shows the dimensionality reduction effect of linear dimensionality reduction LDA.
- ✓ Note that the red and blue dots in the figure are manually typed labels. So we already know the classification information of the sample points.
- ✓ To find a good projection angle so that the sample features in one-dimensional space show small intra-class differences and large inter-class differences.



4.2 Principal Component Analysis (PCA)

Dimensionality reduction.

✓ Feature selection

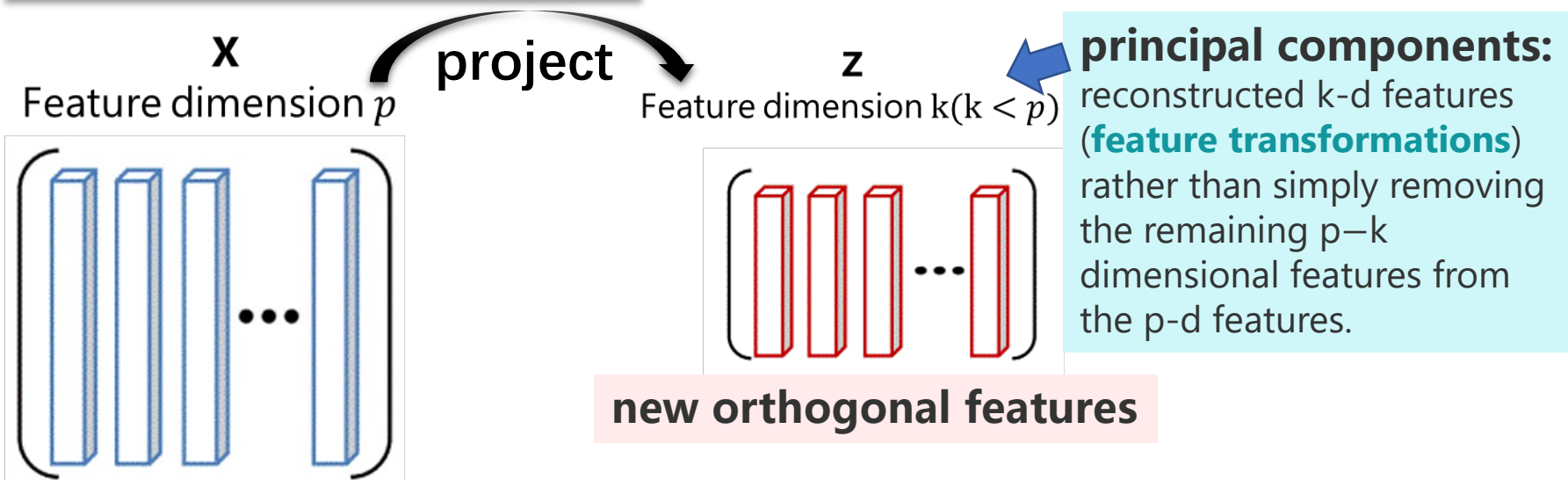
✓ Feature transformation

PCA is the most commonly used linear **dimensionality reduction** method. Its goal is to map high-dimensional data into a low-dimensional space through some linear projection, and **expect the most information (maximum variance) of the data** in the **projected** dimension, thus using fewer data dimensions while more information of original data points.

Feature selection

In PCA, we will perform **feature selection**, mainly to eliminate **features** that are **not related to class labels**. And actually many features are related to class labels, but there is **noise or redundancy** in them. In this case, a feature reduction method is needed to **reduce the number of features**, **reduce noise and redundancy**, and reduce the possibility of **overfitting**.

Feature transformation



4.3 Solving steps of PCA

- 1 Remove the mean
- 2 Compute the covariance matrix
- 3 Compute the eigenvalues and eigenvectors of the covariance matrix
- 4 Sort the eigenvalues
- 5 Keep the eigenvectors corresponding to the top **N** largest eigenvalues
- 6 Transform the original features into the new space constructed by the **N** eigenvectors obtained above

Data=	X_1	X_2	$M = 10$ samples
	2.5	2.4	
	0.5	0.7	
	2.2	2.9	
	1.9	2.2	
	3.1	3.0	
	2.3	2.7	
	2	1.6	
	1	1.1	
	1.5	1.6	
	1.1	0.9	

$N = 2$ features

Dimensionality Reduction

The i th sample $X^i = (x_1^i, x_2^i)^T = (0.5, 0.7)$

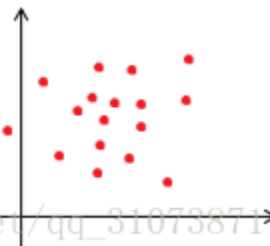
Suppose there are **M** samples $\{X^1, X^2, \dots, X^M\}$, Each sample has **N**-dimensional **features** $X^i = (x_1^i, x_2^i, \dots, x_N^i)^T$, Each feature has its own eigenvalue.

4.3 Step 1: Centering of all features: **de-meaning**

1. Calculate the **mean** of each feature. For all samples, **each feature minus its own mean value**.

Data=

X_1	X_2
2.5	2.4
0.5	0.7
2.2	2.9
1.9	2.2
3.1	3.0
2.3	2.7
2	1.6
1	1.1
1.5	1.6
1.1	0.9

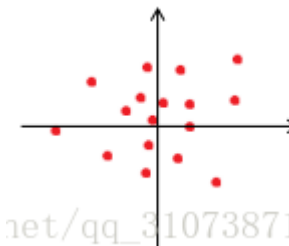


$$\overline{X_1} = 1.81 \quad \overline{X_2} = 1.91$$

Calculate the **mean** of each feature.

DataAdjust=

X_1	X_2
0.69	0.49
-1.31	-1.21
0.39	0.99
0.09	0.29
1.29	1.09
0.49	0.79
0.19	-0.31
-0.81	-0.81
-0.31	-0.31
-0.71	-1.01



each feature **minus** its own mean value.

The step of minus the mean (centering) is not a pre-processing step independent of PCA, but a **mandatory step defined by the definition of PCA**. **We will see the need for it in the derivation of the projection process

4.3 Step 2: Find the **covariance matrix C**

$$2. C = \begin{bmatrix} \text{cov}(x_1, x_1) & \text{cov}(x_1, x_2) \\ \text{cov}(x_2, x_1) & \text{cov}(x_2, x_2) \end{bmatrix}$$

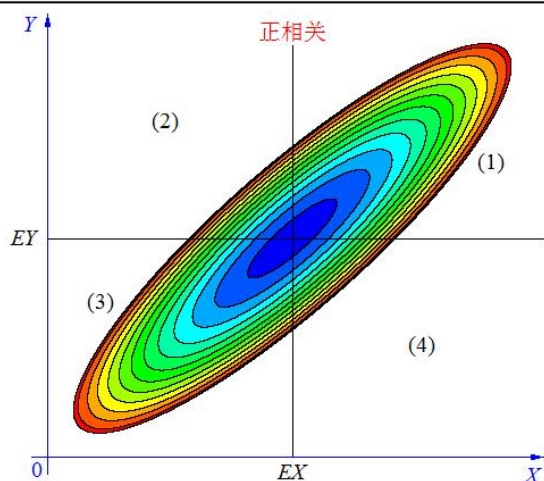
- On the diagonal are the **variances** of the features x_1 and x_2 ;
- Off the diagonal are the **covariances**

Covariance is a **measure** of the **degree of change** in two variables that **change at the same time**.

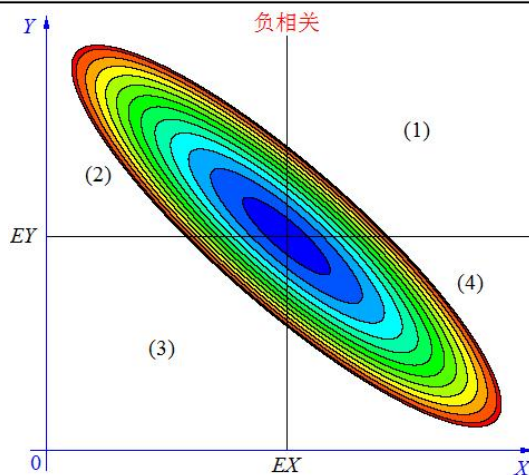
The **larger** the absolute value of the covariance, the **greater** the **influence of the two on each other**, and vice versa, the smaller the influence.

$$\text{cov}(x_1, x_1) = \frac{\sum_{i=1}^M (x_1^i - \bar{X}_1)(x_1^i - \bar{X}_1)}{M - 1}$$
$$\text{cov}(x_2, x_2) = \frac{\sum_{i=1}^M (x_2^i - \bar{X}_2)(x_2^i - \bar{X}_2)}{M - 1}$$

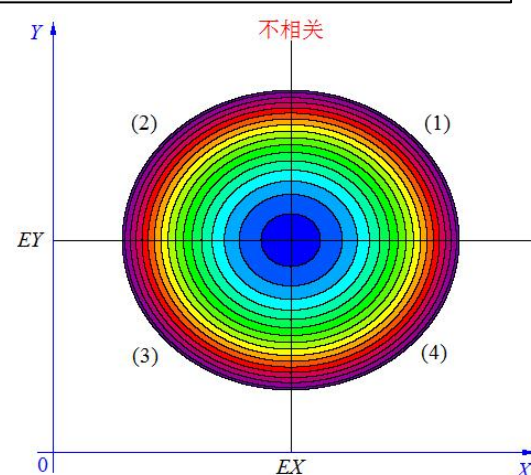
$$\text{cov}(x_1, x_2) = \frac{\sum_{i=1}^M (x_1^i - \bar{X}_1)(x_2^i - \bar{X}_2)}{M - 1}$$
$$\text{cov}(x_2, x_1) = \frac{\sum_{i=1}^M (x_2^i - \bar{X}_2)(x_1^i - \bar{X}_1)}{M - 1}$$



cov > 0: if x_1 or x_2 increase, the other also increases;



cov < 0: x_1 x_2 , one increase, the other decreases;



cov = 0: x_1 and x_2 are not correlated

4.3 Step 2: Find the **covariance matrix C**

$$2. C = \begin{bmatrix} \text{cov}(x_1, x_1) & \text{cov}(x_1, x_2) \\ \text{cov}(x_2, x_1) & \text{cov}(x_2, x_2) \end{bmatrix}$$

$$\text{cov}(x_1, x_1) = \frac{\sum_{i=1}^M (x_1^i - \bar{X}_1)(x_1^i - \bar{X}_1)}{M - 1}$$

$$\text{cov}(x_2, x_2) = \frac{\sum_{i=1}^M (x_2^i - \bar{X}_2)(x_2^i - \bar{X}_2)}{M - 1}$$

$$\text{cov}(x_1, x_2) = \frac{\sum_{i=1}^M (x_1^i - \bar{X}_1)(x_2^i - \bar{X}_2)}{M - 1}$$

$$\text{cov}(x_2, x_1) = \frac{\sum_{i=1}^M (x_2^i - \bar{X}_2)(x_1^i - \bar{X}_1)}{M - 1}$$

DataAdjust=

X_1	X_2
0.69	0.49
-1.31	-1.21
0.39	0.99
0.09	0.29
1.29	1.09
0.49	0.79
0.19	-0.31
-0.81	-0.81
-0.31	-0.31
-0.71	-1.01

$$C = \begin{bmatrix} 0.616555556 & 0.615444444 \\ 0.615444444 & 0.716555556 \end{bmatrix}$$

4.3

Step 3: Find the **eigenvalues** and the **corresponding eigenvectors** of the covariance matrix **C**

3. find the **eigenvalues** λ and the corresponding **eigenvectors** u of the covariance matrix **C**.

C: Real symmetric matrix 实对称矩阵

$$Cu = \lambda u$$

- The number of eigenvalues λ is N (number of features) and each λ_i corresponds to an eigenvector u_i
- Sort the eigenvalues λ **from largest to smallest**
- Select the **top k of the largest eigenvalues** and their corresponding **k eigenvectors** to obtain a set of $\{(\lambda_1, u_1), (\lambda_2, u_2), \dots, (\lambda_k, u_k)\}$
- In this example, the original features are only 2-dimensional, and when selecting λ , let $k = 1$ and choose the largest λ_1 and its corresponding u_1 .

Here the eigenvectors are **normalized, unit vector**

$$C = \begin{bmatrix} 0.616555556 & 0.615444444 \\ 0.615444444 & 0.716555556 \end{bmatrix}$$

$$\begin{aligned} \det(\lambda E - C) &= \begin{vmatrix} \lambda - 0.616555556 & -0.615444444 \\ -0.615444444 & \lambda - 0.716555556 \end{vmatrix} \\ &= (\lambda - 0.616555556)(\lambda - 0.716555556) - 0.615444444^2 = 0 \text{ (实际可以用MATLAB等)} \end{aligned}$$



$$\text{eigenvalues} = \begin{pmatrix} 0.0490833989 \\ 1.28402771 \end{pmatrix}$$

$$\text{eigenvectors} = \begin{pmatrix} -0.735178656 & -0.677873399 \\ 0.677873399 & -0.735178656 \end{pmatrix}$$

For a real symmetric matrix, eigenvectors corresponding to distinct eigenvalues are necessarily orthogonal

4.3

Step 4: Project the original features onto the selected eigenvectors to obtain the new K -dimensional features after **dimensionality reduction**

4. Project the sample points onto the selected eigenvectors.

Suppose the number of samples is m , the number of features is n

the sample matrix after minus the mean is **DataAdjust**($m * n$)

the covariance matrix is $n * n$

the matrix consisting of k eigenvectors is **EigenVectors**($n * k$)

the projected data **FinalData** is **FinalData**($m * k$) = **DataAdjust**($m * n$) $\times$ **EigenVectors**($n * k$)

$$\text{FinalData}(10 * 1) = \text{DataAdjust}(10 * 2) \times \text{EigenVectors}(-0.677873399, -0.735178656)^T$$

DataAdjust=	X_1	X_2	$\times \begin{pmatrix} -0.677873399 \\ -0.735178656 \end{pmatrix} \rightarrow$	<i>FinalData</i>
	0.69	0.49		-0.827970186
	-1.31	-1.21		1.77758033
	0.39	0.99		-0.992197494
	0.09	0.29		-0.274210416
	1.29	1.09		-1.67580142
	0.49	0.79		-0.912949103
	0.19	-0.31		0.991094375
	-0.81	-0.81		1.14457216
	-0.31	-0.31		0.438046137
	-0.71	-1.01		1.22382056

4.3 Mathematical principles of PCA

1. Mathematical definition and geometric meaning of dot product

1) Mathematical definition of dot product

The dot product of two vectors with the same dimension is defined as follows (note that I use row vectors. If column vectors, the transpose sign is marked in the previous vector) **The dot product operation maps two vectors into a real number**

$$(a_1, a_2, \dots, a_n) \cdot (b_1, b_2, \dots, b_n)^T = a_1b_1 + a_2b_2 + \dots + a_nb_n$$

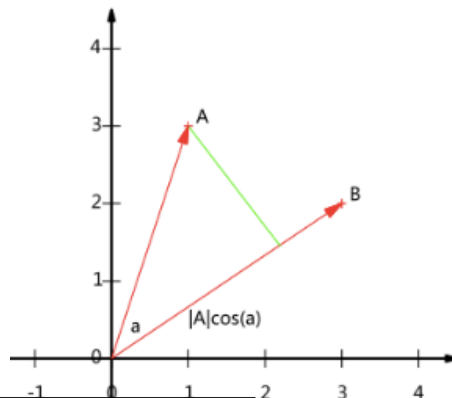
2) Geometric meaning of dot product

An n-dimensional vector can be equivalently represented as a directed line segment emitted from the origin in n-dimensional space.

- There are 2D vectors

$$A = (x_1, y_1), B = (x_2, y_2)$$

- A and B can be represented by two directed line segments starting from the origin



$$A \cdot B = |A||B|\cos(\alpha)$$

projected length
from A to B

When $|B| = 1$

$$A \cdot B = |A|\cos(\alpha)$$

From the definition of dot product, we know the **geometric projection relationship between two vectors**, and the projection represents a **mapping of sample features**.

The dot product value of A and B = the **vector length** of the projection of A to the straight line where B is located

4.3 Mathematical principles of PCA

PS: The main function of dot product

The main function of the dot product is to **calculate angles**

Given the coordinates of two points ***a*** and ***b*** in the three-dimensional *xyz* coordinate system.

How to calculate the angle **θ** between vectors ***a*** and ***b***?

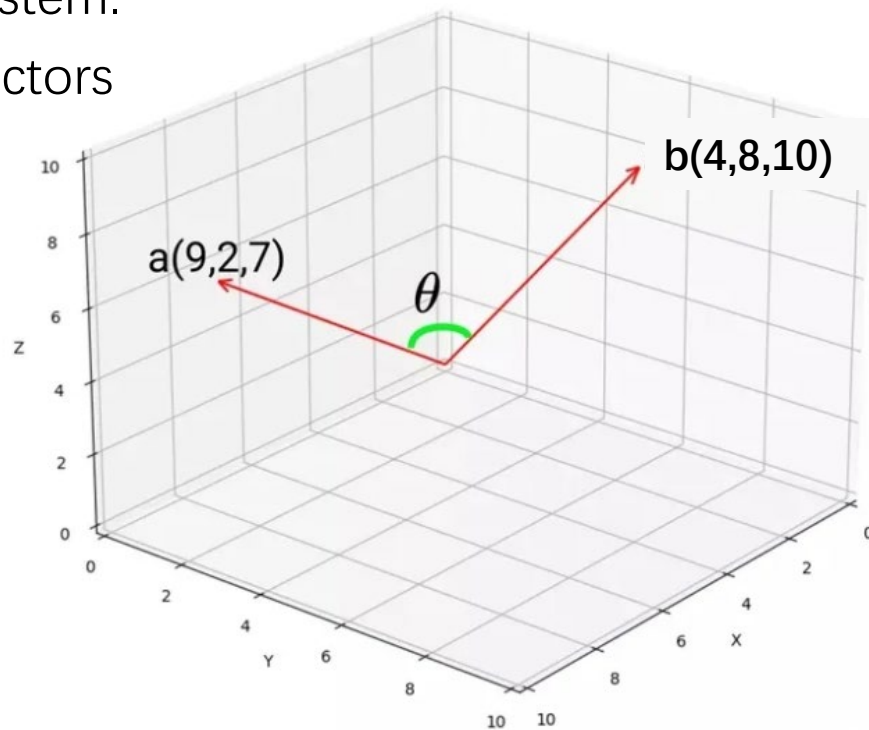
According to the previous dot product formula and geometric definition, we can derive the calculation formula for calculating angle **θ**

$$a \cdot b = |a| \times |b| \times \cos(\theta)$$

$$\cos(\theta) = \frac{a \cdot b}{|a| \times |b|}$$

$$\theta = \cos^{-1} \left(\frac{a \cdot b}{|a| \times |b|} \right)$$

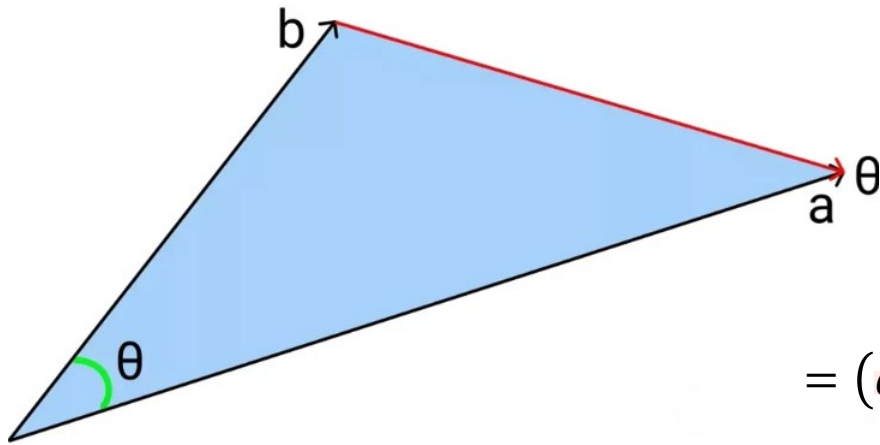
$$= \cos^{-1} \left(\frac{a_1 b_1 + a_2 b_2 + a_3 b_3}{|a| \times |b|} \right) = \cos^{-1} \left(\frac{9 \times 4 + 2 \times 7 + 7 \times 10}{\sqrt{134} \times \sqrt{180}} \right) = \cos^{-1} \left(\frac{122}{\sqrt{134} \times \sqrt{180}} \right)$$



Why the geometric meaning and coordinate definition of dot product are equivalent?

4.3 Mathematical principles of PCA

Why the geometric meaning and coordinate definition of dot product are equivalent?



$$\text{Given } \overrightarrow{BA} = (a_x - b_x, a_y - b_y)$$

$$|BA|^2 = |a|^2 + |b|^2 - 2 \times |a| \times |b| \times \cos \theta$$

$$|BA|^2 = (a_x - b_x)^2 + (a_y - b_y)^2$$

$$= (a_x^2 + a_y^2) + (b_x^2 + b_y^2) - 2 \times |a| \times |b| \times \cos \theta$$

$$|BA|^2 = a_x^2 - 2a_x \cdot b_x + a_y^2 + b_x^2 - 2a_y \cdot b_y + b_y^2$$

$$= a_x^2 + a_y^2 + b_x^2 + b_y^2 - 2 \times |a| \times |b| \times \cos \theta$$

$$-2a_x \cdot b_x - 2a_y \cdot b_y = -2 \times |a| \times |b| \times \cos \theta$$

Mathematical definition $a_x \cdot b_x + a_y \cdot b_y = |a| \times |b| \times \cos \theta$ Geometric meaning

equivalent

$\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}||\mathbf{B}| \cos(\alpha) \Rightarrow$ When $|\mathbf{B}|=1$, the dot product value of A and B = the vector length of the projection of A to the straight line where B is located.

4.3 Mathematical principles of PCA

2. Basis: the projection axis B we choose

A two-dimensional vector can correspond to a **directed line segment starting from the origin** in the two-dimensional Cartesian coordinate system.

For example, the vector on the right can be expressed as **(3,2)**

However, we often overlook that only one (3,2) itself cannot accurately represent a vector.

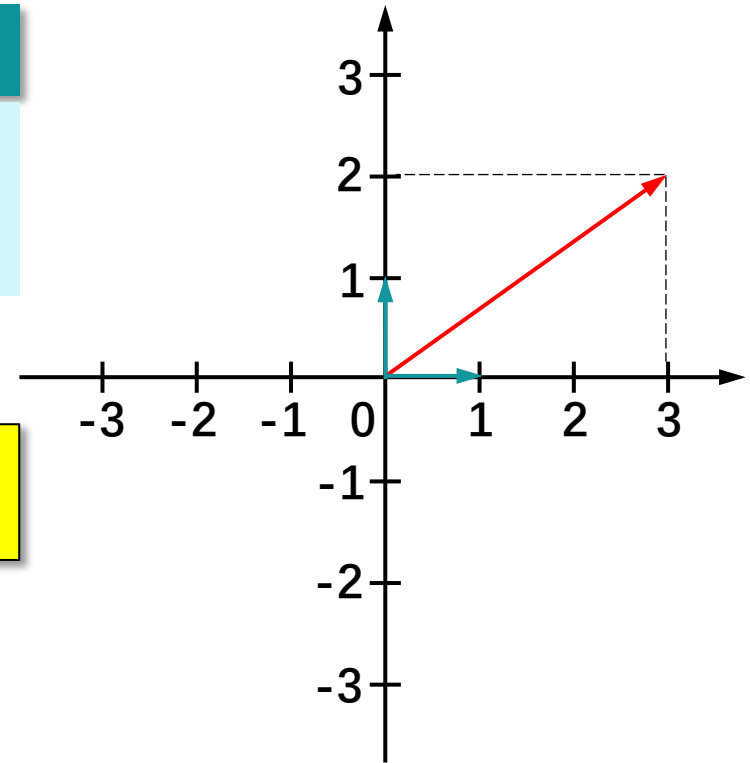
The coordinates (3,2) here actually represent the projection value 3 of the vector on the x-axis. The projected value on the y-axis is 2.

Taking **the vector with length 1 in the positive direction on the x-axis and y-axis as the standard**, then a vector (3,2) actually means that the projection **on the x-axis** is 3 and the projection **on the y-axis** is 2.

The vector (x,y) actually represents a linear combination:

$$x(1, 0)^T + y(0, 1)^T$$

(1,0) and (0,1) are called a set of basis of two-dimensional space



Note that the projection is a vector, so can be negative.

4.3 Mathematical principles of PCA

In fact, any two linearly independent two-dimensional vectors can become a set of basis.

Linear independence can be viewed as two vectors that are not on a straight line in a two-dimensional plane.

We hope that the modulus of the basis is 1. According to the definition of dot product, we can directly obtain the coordinates of the vector on the new basis by dot product them.

$$A \cdot B = |A| \cos(\alpha)$$

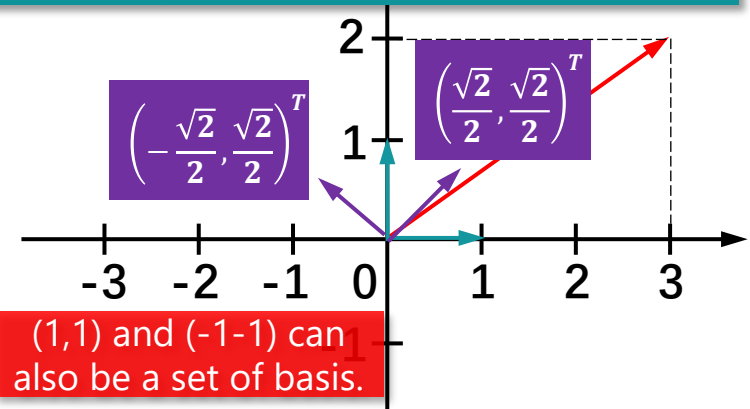
For any vector $(1,1)^T$, by dividing it by its **modulus** ($\sqrt{2}$), can find a unit vector in the same direction.

Now we want to obtain the coordinates of $(3,2)$ on the new basis, which means the projected vector values in two directions. According to the geometric meaning of the dot product, we only need to calculate the dot product between $(3,2)$ and two basis respectively.

The inner product of $(3,2)$ and $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)^T = 3 \times \frac{\sqrt{2}}{2} + 2 \times \frac{\sqrt{2}}{2} = \frac{5\sqrt{2}}{2}$,

The new coordinate of the red line on the new basis 1 is $\frac{5\sqrt{2}}{2}$

2. we choose the projection axis B (Basis)



In the examples we gave, the basis is **orthogonal** (i.e., the inner product is zero, or they are **perpendicular** to each other). However, the only requirement for a set of vectors to form a basis is linear independence. Still, because orthogonal bases have nice properties, the bases we generally use are orthogonal.

The inner product of $(3,2)$ and $\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)^T = -3 \times \frac{\sqrt{2}}{2} + 2 \times \frac{\sqrt{2}}{2} = -\frac{\sqrt{2}}{2}$,

The new coordinate of the red line on the new basis 2 is $-\frac{\sqrt{2}}{2}$

4.3 Mathematical principles of PCA

3. Matrix representation of basis transformation

Recall: If you want to transform (3,2) into coordinates on a new basis, **use (3,2) to do the dot product operation** with the first new basis as the first new coordinate component, and then use (3, 2) Perform an inner product operation with the second new basis as a component of the second new coordinate.

? If we have many points that need to perform similar dot product operations and calculate their coordinates on the new basis, how should we deal with them?

From vector to matrix

New Basis

$$\begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix} \begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 5/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}$$

coordinates on the new basis

The first point coordinates

$$\begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_R \end{pmatrix} (a_1 \quad a_2 \quad \cdots \quad a_M) = \begin{pmatrix} p_1 a_1 & p_1 a_2 & \cdots & p_1 a_M \\ p_2 a_1 & p_2 a_2 & \cdots & p_2 a_M \\ \vdots & \vdots & \ddots & \vdots \\ p_R a_1 & p_R a_2 & \cdots & p_R a_M \end{pmatrix}$$

4.3 Mathematical principles of PCA

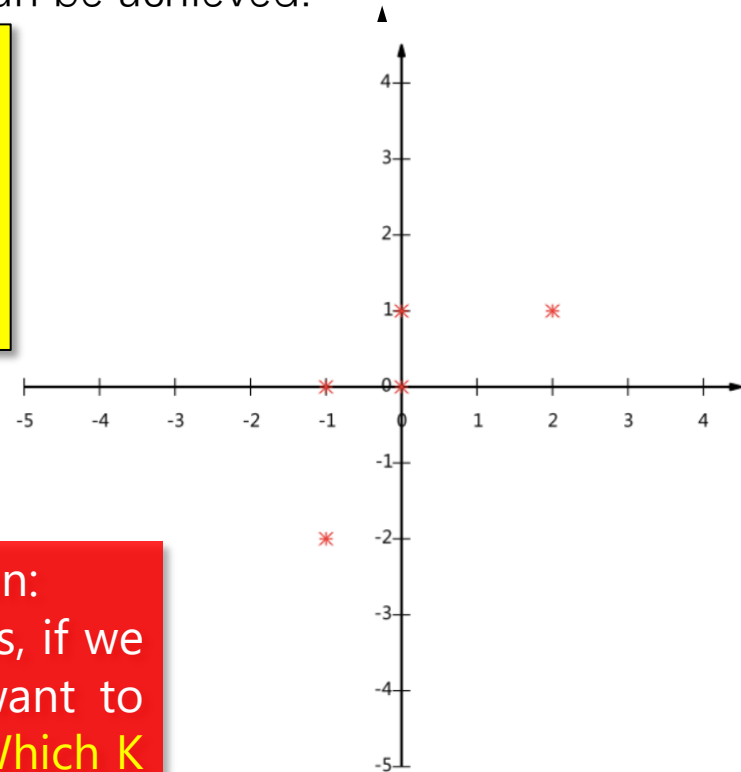
4. Covariance matrix and optimization goals: finding the optimal basis

We discussed that **choosing different basis can give different representations of the same set of data**, and if **the number of basis is less than the dimension of the vector** itself, the effect of dimensionality reduction can be achieved.

For example, if we map the two-dimensional coordinate x onto a one-dimensional basis, then the point $(3,2)$ from the previous page will be mapped onto the new basis $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$, resulting in $\frac{5\sqrt{2}}{2}$. The number of basis vectors $R=1$, the original dimension of the vector $N=2$, and the resulting new coordinate dimension after mapping is 1, so dimensionality reduction is achieved.

$$\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} \frac{5\sqrt{2}}{2} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}$$

We have not yet answered the most critical question: **how to choose the optimal basis**, or in other words, if we have a set of N -dimensional vectors and now want to reduce them to K -dimensional (K is less than N). **Which K basis should we choose to retain the original information to the greatest extent?**



The projected values after projection are as **dispersed** as possible

4.3 Mathematical principles of PCA

5. Variance: a measure of the dispersion of projected values

$$\text{Var}(a) = \frac{1}{m} \sum_{i=1}^m (a_i - \mu)^2$$

$$\text{Var}(a) = \frac{1}{m} \sum_{i=1}^m a_i^2$$

Find a one-dimensional basis that **maximizes the variance** after **all data** is transformed into coordinate representation on this basis.

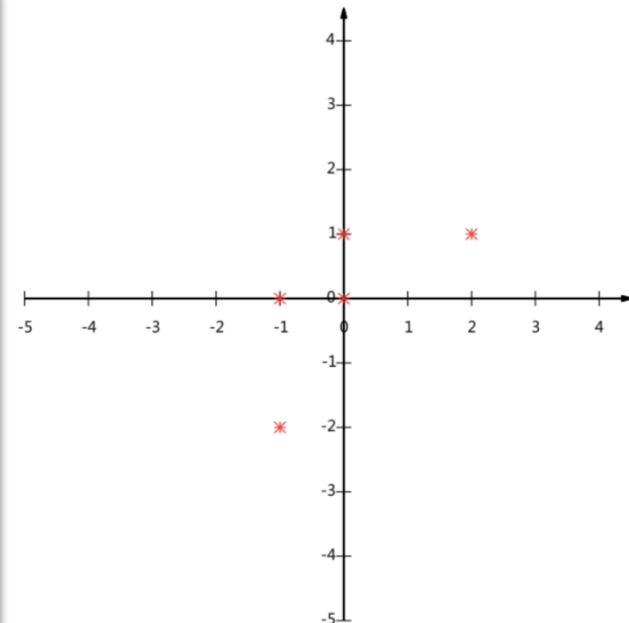
6. Covariance

?

For the above problem of reducing two dimensions to one dimension, just **find the direction that maximizes the variance**.

However, for higher dimensions, there is still a problem that needs to be solved: Consider the problem of reducing three dimensions to two dimensions, first we hope to find a direction that maximizes the variance after projection. **Then we Choose a second projection direction**.

If we simply choose the direction with the largest variance, obviously this direction and the first direction should be **"almost coincident"**. Obviously such dimensions are useless, so there should be **other constraints**.



4.3 Mathematical principles of PCA

6. Covariance:

let the two **new projection fields** represent as much original information as possible. We do **not want a linear correlation** between them, because the correlation means that the two fields are **not completely independent**, and there must be **repeated representation of information**.

$$\text{Cov}(a, b) = \frac{1}{m} \sum_{i=1}^m a_i b_i$$

The covariance of two fields is used to express their correlation.

The covariance of two fields is expressed concisely as its **dot product** divided by the number of elements m .

When the covariance is 0, it means that the two fields are completely independent.

In order to make the **covariance=0**, we can only choose the second one in the direction **orthogonal** to the first basis. Therefore, the two directions finally selected must be **orthogonal** 正交.

$$\begin{aligned} \text{cov}(x_1, x_1) &= \frac{\sum_{i=1}^M (x_1^i - \bar{X}_1)(x_1^i - \bar{X}_1)}{M - 1} \\ \text{cov}(x_2, x_2) &= \frac{\sum_{i=1}^M (x_2^i - \bar{X}_2)(x_2^i - \bar{X}_2)}{M - 1} \end{aligned}$$

$$\begin{aligned} \text{cov}(x_1, x_2) &= \frac{\sum_{i=1}^M (x_1^i - \bar{X}_1)(x_2^i - \bar{X}_2)}{M - 1} \\ \text{cov}(x_2, x_1) &= \frac{\sum_{i=1}^M (x_2^i - \bar{X}_2)(x_1^i - \bar{X}_1)}{M - 1} \end{aligned}$$

Objective: In a symmetric covariance matrix, all off-diagonal elements are zero.

4.3 Mathematical principles of PCA

6. Covariance: covariance between any two elements is zero, variance of each element is as large as possible

We have obtained the **optimization objective** for the **dimensionality reduction** problem: to reduce a set of N-dimensional vectors to **k dimensions** ($k > 0$ and $k < N$), the goal is to select **k orthogonal unit vectors** (i.e., a **basis with modulus 1**) such that after transforming the original data onto this basis, the **covariance between any two elements is zero**, and the **variance of each element is as large as possible** (under the orthogonality constraint, take the largest k variances)

$$X = \begin{pmatrix} a_1 & a_2 & \cdots & a_m \\ b_1 & b_2 & \cdots & b_m \end{pmatrix}$$
$$\frac{1}{m} X X^T = \begin{pmatrix} \frac{1}{m} \sum_{i=1}^m a_i^2 & \frac{1}{m} \sum_{i=1}^m a_i b_i \\ \frac{1}{m} \sum_{i=1}^m a_i b_i & \frac{1}{m} \sum_{i=1}^m b_i^2 \end{pmatrix}$$

covariance between any two elements is zero

variance of each element is as large as possible

Suppose we have m n-dimensional data records, and we arrange them column-wise into an $n \times m$ matrix X. Let $C = 1/m * X * X^T$ Then C is a symmetric matrix, whose diagonal entries are the variances of the individual fields, and the element in the i-th row and j-th column (which equals the element in the j-th row and i-th column) represents the covariance between the i-th and j-th fields.

We have the optimization objective — but what about the action plan?

4.3 Mathematical principles of PCA

6. Covariance:

Optimization objective: diagonalize the covariance matrix, set all off-diagonal elements to zero, and arrange the diagonal elements in descending order from top to bottom.

Let the covariance matrix of the original data matrix X be C , and let P be a matrix formed by a set of basis vectors arranged in rows. Let $Y = PX$. Then Y is the data after applying the basis transformation P to X . Let the covariance matrix of Y be D .

$$\begin{aligned} D &= \frac{1}{m} Y Y^T \\ &= \frac{1}{m} (PX)(PX)^T \\ &= \frac{1}{m} P X X^T P^T \\ &= P \left(\frac{1}{m} X X^T \right) P^T \\ &= P C P^T \end{aligned}$$

D : In the new covariance matrix, all off-diagonal elements are set to zero, and the diagonal elements are arranged from top to bottom in descending order — diagonalization.

P : can diagonalize the original covariance matrix C

Optimization objective: Find a matrix P such that PCP^T is a diagonal matrix, with the diagonal elements arranged in descending order from largest to smallest.

Then the **first K rows of P are the basis** we are looking for. Multiplying the matrix formed by the first K rows of P by X reduces X from N dimensions to K dimensions while satisfying the above optimization conditions.

4.3 Mathematical principles of PCA

6. Covariance:

- ◆ The covariance matrix C is a symmetric matrix
- ✓ For a real symmetric matrix, eigenvectors corresponding to distinct eigenvalues are necessarily orthogonal.
- ✓ If an eigenvalue has multiplicity r , then there must exist r linearly independent eigenvectors corresponding to that eigenvalue, and thus these r eigenvectors can be orthonormalized.

An $n \times n$ real symmetric matrix always has n orthonormal eigenvectors.

$$E = (e_1 \quad e_2 \quad \cdots \quad e_n)$$

$$E^T C E = \Lambda = \begin{pmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{pmatrix}$$

$$P = E^T$$

P is the matrix formed by arranging the **orthonormal** eigenvectors of the covariance matrix **in rows** (note the orientation), where each row of **P** is an eigenvector of **C**. If **P** arranges the eigenvectors from top to bottom according to the descending order of the eigenvalues in Λ , then multiplying the matrix consisting of the first K rows of **P** by the original data matrix **X** yields the reduced-dimensional data matrix **Y** that we need.

4.4

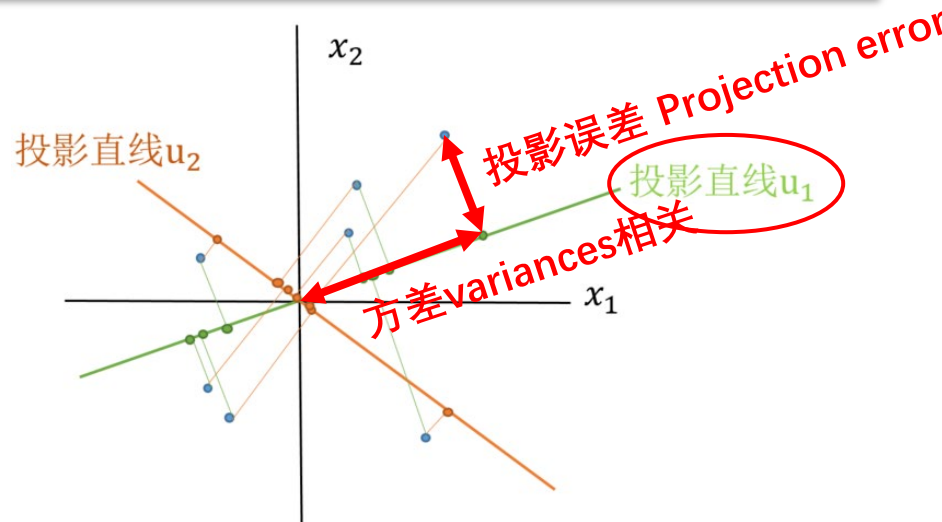
Why eigenvectors are projection plane, eigenvalues are the variances

?

Why can the eigenvectors of the covariance matrix be regarded as the projection plane, and the corresponding eigenvalues are the variances of the original features after it has been projected onto this projection plane?

- ✓ After minus the mean and centralizing all features

Maximum Variance Theory: In signal processing, it is believed that the signal has a larger variance while the noise has a smaller variance; the larger the variance, the greater the amount of information. Each eigenvector of the **covariance matrix** represents a projection plane, and the eigenvalue corresponding to each eigenvector is the variance of the original features projected onto that plane. Since we want to preserve as much information as possible after projection, we should choose the projection plane with a larger variance to project the original features — that is, select the eigenvectors corresponding to **larger eigenvalues**.



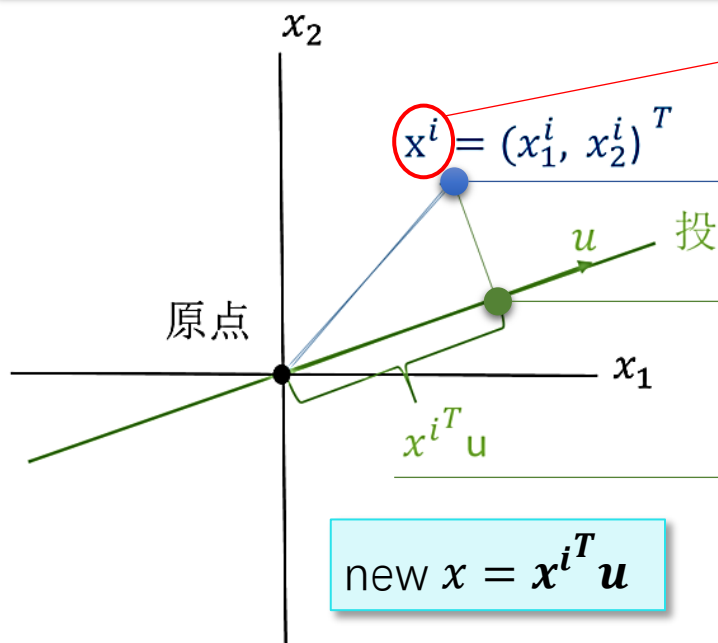
- ✓ Project these samples to the low-dimensional space
- ? There are many projection line, such as the following u_1 and u_2 , choose which one is better?

theory of **maximum variance**

the projection line with the **larger** sample **variance** after projecting over is chosen as the projection dimension

4.4 Why eigenvectors are projection plane, eigenvalues are the variances

how is the sample variance calculated after projection?



Taking a sample X^i as an example

the blue points indicate that sample, the original feature has 2 dimensions

投影直线 $u = (u_1^i, u_2^i)^T$

the green points indicate the projection of the sample on u , and u is the **unit vector** of that projected line

After projection, X^i lies on the eigenvectors. We need to identify **larger** new one-dimensional **features** (which retain more information), corresponding to larger stretching factors (**eigenvalues**).

Then the original two-dimensional feature of this sample $X^i = (x_1, x_2)^T$ projected onto this projection line u becomes a one-dimensional feature (changed from a two-dimensional coordinate system to a one-dimensional coordinate system), and the **value of this one-dimensional feature = the distance from the projection point to the origin: $x^iT u$** (according to the formula of vectors). [Geometry: vector dot product]

The other samples are projected onto the projection line, and the new 1D features generated are calculated in the same way.

variance

According to the definition of variance, the variance between all samples after projection is equal to

$$\text{variance} = \frac{1}{M} \sum_{i=1}^M (x^iT u - \text{The mean of all } x^iT u)^2 = \frac{1}{M} \sum_{i=1}^M (x^iT u)^2$$

Note: The term 'The mean of all $x^iT u$ ' is circled in red in the original image, with a red '=0?' written above it.

FinalData

-0.827970186
1.77758033
-0.992197494
-0.274210416
-1.67580142
-0.912949103
0.991094375
1.14457216
0.438046137
1.22382056

PS Unbiased estimation



Do you remember the denominator $M - 1$ in the covariance matrix? The variance is a special case of covariance, and we mentioned then that **$M - 1$ in the denominator is an unbiased estimate of the sample.**

And in most cases, we do not know the mathematical expectation μ of the sample x , so **the mean of the sample $\bar{x}$ is used instead of μ** , but **$\bar{x}$ and μ are not equal**, and if the denominator is M , the **variance is underestimated.**

$$\begin{aligned} cov(x_1, x_1) &= \frac{\sum_{i=1}^M (x_1^i - \bar{X}_1)(x_1^i - \bar{X}_1)}{M - 1} \\ cov(x_2, x_2) &= \frac{\sum_{i=1}^M (x_2^i - \bar{X}_2)(x_2^i - \bar{X}_2)}{M - 1} \end{aligned}$$

Assuming that the denominators of both the variance and covariance are M , the variance is of the following form

$$\begin{aligned} \frac{1}{M} \sum_{i=1}^M (x_i - \bar{x})^2 &= \frac{1}{M} \sum_{i=1}^M [(x_i - \mu) + (\mu - \bar{x})]^2 \\ &= \frac{1}{M} \sum_{i=1}^M (x_i - \mu)^2 + \frac{2}{M} \sum_{i=1}^M (x_i - \mu)(\mu - \bar{x}) + \frac{1}{M} \sum_{i=1}^M (\mu - \bar{x})^2 \\ &= \frac{1}{M} \sum_{i=1}^M (x_i - \mu)^2 + 2(\bar{x} - \mu)(\mu - \bar{x}) + (\mu - \bar{x})^2 = \frac{1}{M} \sum_{i=1}^M (x_i - \mu)^2 \quad \text{[variance]} - (\mu - \bar{x})^2 \end{aligned}$$

However, when the sample size is large, the difference between $M-1$ and M is not significant, so $M-1$ can be changed to M to easily calculate

Obviously, $\frac{1}{M} \sum_{i=1}^M (x_i - \bar{x})^2 < \frac{1}{M} \sum_{i=1}^M (x_i - \mu)^2$ (variance) **unless $\bar{x} = \mu$** . So we use **$M - 1$** to **scales up the originally small estimates**, which could converges to the variance and covariance.

4.4 Why eigenvectors are projection plane, eigenvalues are the variances

variance

According to the definition of variance, the variance between all samples after projection is equal to

$$\text{variance} = \frac{1}{M} \sum_{i=1}^M (x^i T u - \text{所有 } x^i T u \text{ 的均值})^2 = \frac{1}{M} \sum_{i=1}^M (x^i T u)^2$$

Since each feature of these sample undergoes a **de-meaning operation** at the very beginning, **the mean value of each dimensional feature of x^i is 0**, so the **mean value of the features projected onto u remains 0**.

$$\text{variance} = \frac{1}{M} \sum_{i=1}^M \underset{\text{scalar}}{(x^i T u)^2} = \frac{1}{M} \sum_{i=1}^M \underset{\text{scalar}}{u^T x^i x^i T u} = u^T \left(\frac{1}{M} \sum_{i=1}^M x^i x^i T \right) u$$

Quadratic form, of the form $x^T A x$, A is a symmetric square matrix; think of the covariance matrix C .

If $X = [x_1 \ x_2 \ \cdots \ x_M]$, $X^T = \begin{bmatrix} x_1^T \\ x_2^T \\ \vdots \\ x_M^T \end{bmatrix}$ $\rightarrow XX^T = \sum_{i=1}^M x^i x^i T$, so **variance** = $\frac{1}{M} u^T XX^T u$

Assuming that a certain eigenvalue of XX^T is λ , and the corresponding eigenvector is ξ :

$$(XX^T)^T = XX^T, \quad XX^T \xi = \lambda \xi \rightarrow (XX^T \xi)^T \xi = (\lambda \xi)^T \xi \rightarrow \xi^T XX^T \xi = \lambda \xi^T \xi \rightarrow$$

$$\xi^T XX^T \xi = (X^T \xi)^T (X^T \xi) = \|X^T \xi\|^2 = \lambda \xi^T \xi = \lambda \|\xi\|^2 \geq 0$$

4.4 Why eigenvectors are projection plane, eigenvalues are the variances

Since $\lambda \|\xi\|^2 \geq 0$, so $\lambda \geq 0 \rightarrow \mathbf{X}\mathbf{X}^T$ is semi-positive definite symmetric matrix $\rightarrow \mathbf{u}^T \mathbf{X}\mathbf{X}^T \mathbf{u}$ is the quadratic form of semi-positive definite array.

In PCA, **centering and selecting the eigenvector corresponding to the largest eigenvalue** as the projection axis are key steps to ensure the maximization of variance after projection. The reasons are as follows:

1. Why is centering necessary?

The goal of PCA is to find the direction that maximizes the variance after projection, and variance is the deviation from the mean. If the data are not centered (i.e., the mean is not subtracted), the computed covariance matrix will be contaminated by the effect of the mean, causing the principal components to be biased toward the overall offset direction of the data (i.e., the first principal component will roughly point in the direction of the mean) rather than the direction of greatest variation within the data.

After centering, each feature has a mean of zero, and the covariance matrix

$$\mathbf{C} = \frac{1}{M} \sum_{i=1}^M \mathbf{x}^i \mathbf{x}^{iT}$$

directly reflects the spread of the data around the origin. At this point, maximizing $\mathbf{u}^T \mathbf{C} \mathbf{u}$ is equivalent to **maximizing the variance after projection** (because the mean after projection is also zero).

2. Why select the eigenvector corresponding to the largest eigenvalue?

The projection variance can be expressed as $\text{var} = \mathbf{u}^T \mathbf{C} \mathbf{u}$ under the constraint $\|\mathbf{u}\| = 1$.

By the **Rayleigh quotient theorem**: for a real symmetric matrix $\mathbf{C}$,

$$\max_{\|\mathbf{u}\|=1} \mathbf{u}^T \mathbf{C} \mathbf{u} = \lambda_{\max},$$

and the maximum is achieved when $\mathbf{u}$ is the eigenvector corresponding to $\lambda_{\max}$.

The eigenvalue λ represents the variance of the data projected onto the direction of the corresponding eigenvector. Choosing the eigenvector associated with the largest eigenvalue therefore selects the projection direction that retains the most information (largest variance).

3. Why not other directions?

For any other direction $\mathbf{v}$, the projection variance $\mathbf{v}^T \mathbf{C} \mathbf{v}$ is less than or equal to the largest eigenvalue. By spectral decomposition, $\mathbf{v}$ can be expressed as a linear combination of eigenvectors, and its variance is a weighted sum of the eigenvalues. The maximum value is clearly achieved only by the direction corresponding to the largest eigenvalue.

Summary

- Centering ensures that the covariance matrix correctly reflects the spread of the data (rather than the mean shift).
- The eigenvector corresponding to the largest eigenvalue gives the direction that maximizes the variance after projection, i.e., the first principal component.
- Subsequent principal components (eigenvectors corresponding to the next largest eigenvalues, orthogonal to the already chosen ones) successively maximize the remaining variance under the orthogonality constraint.

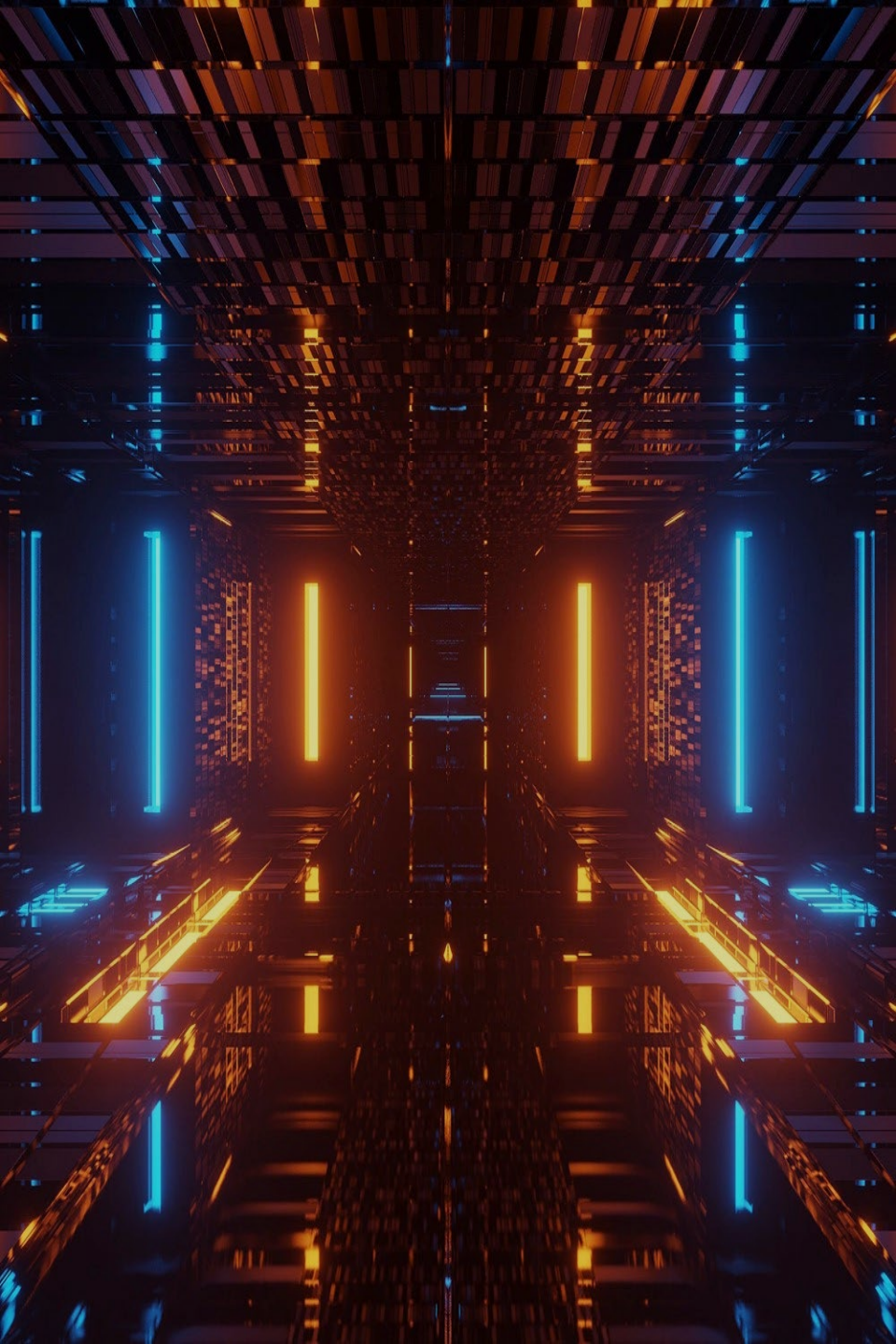
4.5 Advantages and disadvantages of PCA

Advantages

1. Unsupervised learning, unrestricted by sample labels, measures the degree of information retention in terms of variance.
2. Since the **covariance matrix** is **symmetric**, the **k eigenvectors are orthogonal** to each other, i.e., the **principal components are orthogonal** to each other, and the orthogonality is definitely **linearly uncorrelated**, which can **eliminate the mutual influence** between the components of the original data.
3. It can reduce the workload of indicator selection.
4. **Using a few indicators** instead of most indicators, using PCA to **reduce the dimensionality** is the most common algorithm.
5. The calculation method is simple and easy to implement on the computer.

Disadvantages

1. Principal components **explain their meaning** often with some **vagueness**, less complete than the original sample.
2. Principal components with **small contribution rates may often contain important information** about the differences in the samples, that is, they may be more useful for distinguishing the categories (labels) of the samples.
3. Whether the orthogonal vector space of the eigenvalue matrix is unique is open to discussion.
4. unsupervised learning.



Thanks